

Final Project — Progress Report

Multi-Faceted Customer Intelligence for E-commerce An End-to-End Data Mining Pipeline on Online Retail II

Data Mining 2026 — Team 4, University of Information Technology, VNU-HCM
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1. Problem Definition

Online retailers routinely ask three separate questions about their customer base: who their customers are (segmentation), which customers are about to stop buying (churn), and which products tend to be bought together (market-basket patterns). In practice these questions are usually tackled in isolation, which misses the connections between them — for example, a retention campaign is far more effective when the offer matches the purchasing pattern of the customer's segment.

For our final project we plan to build a single, coherent data-mining pipeline that addresses all three questions on the same dataset and then combines their outputs into a cross-task analysis (e.g., churn rate per segment and the basket patterns that characterise each segment). The practical motivation is well established: retaining an existing customer is substantially cheaper than acquiring a new one, so being able to identify which customers will churn — and why, in the context of their segment and baskets — supports concrete, targeted retention decisions.

Concretely, the project aims to (i) segment customers from their purchase behaviour, (ii) predict 90-day churn, (iii) mine product association rules, and (iv) synthesise these results into segment-aware business recommendations.

2. Dataset

We use the Online Retail II dataset from the UCI Machine Learning Repository (dataset ID 502, <https://archive.ics.uci.edu/dataset/502/online+retail+ii>). It contains real transaction records from a UK-based online retailer between December 2009 and December 2011 — about 1.07 million invoice-line records, roughly 5,900 unique customers and 5,300 stock items, spread across 41 countries (predominantly the United Kingdom).

Each record has the following main fields:

- Invoice — transaction/invoice number (a leading 'C' marks a cancellation).
- StockCode and Description — product identifier and text label.
- Quantity and Price — units purchased and unit price.
- InvoiceDate — timestamp of the transaction.

- Customer ID — anonymised customer identifier (missing on ~25% of rows).
- Country — customer's country.

From these raw fields we plan to engineer per-customer features: the classic RFM variables (Recency, Frequency, Monetary) together with simple behavioural features such as average basket value, average basket size, number of unique products purchased, and the customer's dominant country. These engineered features will be the common input shared across the segmentation and churn tasks.

3. Intended Methodology

Our planned pipeline covers preprocessing and four mining tasks; the main steps are outlined briefly below.

Preprocessing and feature engineering.

We will clean the data by removing rows with missing Customer ID, dropping cancellation invoices and non-positive Quantity/Price values, and winsorising extreme Quantity/Price values. We then derive the RFM and behavioural features above, and construct a churn label using a temporal cutoff date with a fixed 90-day post-cutoff observation window (a customer active before the cutoff but absent during the window is labelled churned).

Customer segmentation (clustering).

We intend to compare three clustering families — K-means (partitional), DBSCAN (density-based), and agglomerative/Ward hierarchical clustering — on log-transformed, standardised RFM features. Cluster quality will be assessed with internal validity indices (Silhouette, Davies-Bouldin, Calinski-Harabasz), and the segments will be visualised in 2-D using PCA and t-SNE. We also plan to check that the chosen segmentation is stable when K-means is re-run across several random seeds.

Churn classification.

We plan to train several classical classifiers (e.g., Logistic Regression, Decision Tree, Random Forest, k-NN, SVM, and Naive Bayes) and a small neural network (MLP), using stratified cross-validation plus a held-out test set, and addressing class imbalance with SMOTE applied to the training data only. Models will be compared on AUC, F1, precision and recall. To put those scores in context we also intend to evaluate simple baselines (a majority-class predictor and a recency-only model), run a feature-group ablation, and check the calibration of the predicted probabilities. A key design concern we are paying attention to is avoiding target leakage when building the churn features (computing them only from data before the cutoff).

Association rule mining.

We will mine frequent itemsets and association rules over invoice-level baskets — restricted to frequently-purchased items to keep mining tractable — using both the Apriori and FP-Growth

algorithms, ranking rules by support, confidence and lift, and comparing the two algorithms' runtimes. We will also mine rules within individual segments for the cross-task analysis.

Deep-learning extension and cross-task synthesis.

As an extension we plan to add an autoencoder for unsupervised detection of anomalous customers (via reconstruction error) and to use SHAP to interpret the churn model's drivers. Finally, the cross-task step will join the segmentation, churn, and association-rule results to produce per-segment churn rates and per-segment basket patterns — the analysis we expect to yield the most useful business insight.

Throughout, we aim to keep the work reproducible (fixed random seed, a documented run procedure, and shared utility code) so that all reported numbers can be regenerated.

4. Progress to Date and Remaining Work

Completed so far:

- Project setup: repository structure, environment, and shared utilities.
- Dataset acquired and loaded; initial cleaning and exploratory data analysis carried out (the cleaning step reduces the ~1.07M raw rows to roughly 0.8M usable transactions).
- RFM and behavioural feature engineering implemented; first clustering experiments run.

In progress:

- Churn classification (classical models and the MLP) and association-rule mining.
- Refining the churn feature construction to ensure it is leakage-free.

Remaining work before the final submission:

- Deep-learning extension (autoencoder + SHAP) and the cross-task synthesis.
- Consolidating evaluation tables and figures, then writing the final IEEE-format report and slide deck.